

AUTOMATED DETECTION OF SATELLITE TRAILS IN GROUND-BASED OBSERVATIONS

U-Net segmentation with probabilistic Hough post-processing

Baron Gracias | MPhil Data Intensive Science | University of Cambridge | Supervisor: Dr Eduardo Gonzalez-Solares

Motivation & Scientific Justification

Low-Earth-orbit satellite constellations increasingly cross wide-field astronomical exposures, leaving bright linear trails that corrupt photometry, astrometry, and transient-source catalogues. Manual inspection does not scale to survey operation, so observatories need pixel-reliable automated masks that can run inside reduction pipelines.

This project replicates the satellite-trail detector of Stoppa et al. (2024): a U-Net trained on labelled MeerLICHT image patches, followed by a probabilistic Hough transform that reconnects linear trail evidence missed by the network. The replication is used diagnostically: it tests where the residual error comes from, not only whether a headline score reproduces.

Key Findings

The replication reproduces the target paper’s recall closely: five-seed sampled-test recall is 0.923 ± 0.007 against the paper’s reported 0.94. Strict precision is lower, at 0.847 ± 0.010 on the sampled test set and 0.780 on the parity set that restores the natural negative-patch prevalence. This shows that precision is prevalence-sensitive, whereas recall is the cleaner like-for-like comparison across the two test populations. Image-cluster uncertainty is the relevant interval because source images, not patches, are the independent units.

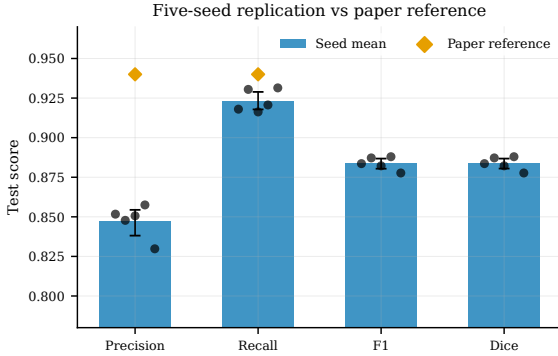


Figure 1: Five-seed replication performance for the locked U-Net compared with Stoppa et al. Paper-reference markers are shown only for the explicitly reported precision and recall values; recall is closely reproduced while strict precision remains lower, motivating the boundary-near error analysis.

The Hough stage reproduces the paper’s intended completeness behaviour. On the full-coverage tiled canvas, trail-pixel recall rises from 0.918 to 0.990, equivalently reducing the pixel-level false-negative rate from 0.082 to 0.0105. This does not make the Hough mask a better strict pixel classifier: post-Hough line drawing increases false-positive area, reducing strict precision from 0.780 to 0.410 on the parity canvas.

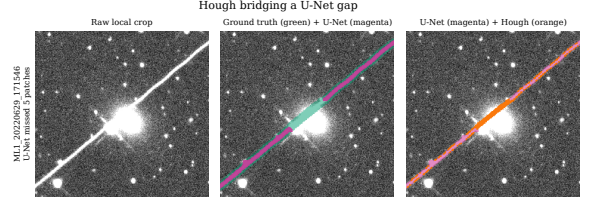


Figure 2: Hough bridging a U-Net gap over a bright star: linear post-processing recovers ground-truth trail pixels missed by the thresholded U-Net.

Diagnostics place much of the strict precision gap near annotated boundaries. False-positive decomposition shows that only 17.5% of false-positive pixels occur in trail-free patches; the remaining 82.5% fall within trail-bearing patches. A one-pixel boundary tolerance raises five-seed precision from 0.847 to 0.946, and 57% of all false positives lie within one pixel of a labelled trail. These diagnostics are consistent with residual error concentrated near labelled trail boundaries. A blinded single-author re-annotation of 64 sealed crops is consistent with this: the original masks, the re-annotation, and the model all share a median width of 6 px, and the model agrees with the re-annotation about as well as the original labels do, with no strict- F_1 difference resolved under this design (paired -0.0001 , 95% CI spanning zero). This is a single-author comparison between mask sources, not inter-annotator agreement. A small over-firing residual remains on adversarial controls, so the audit supports a boundary-scale interpretation without implying a uniformly thin label or model over-paint.

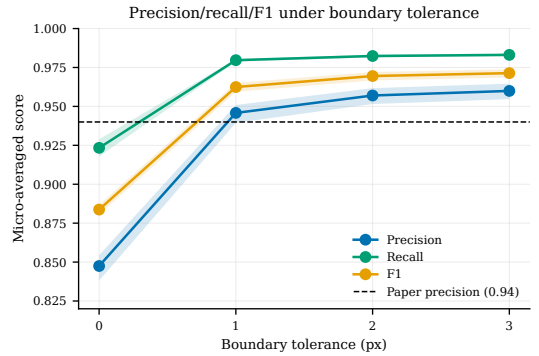


Figure 3: A one-pixel tolerance largely closes the strict precision gap, consistent with residual disagreement concentrated near labelled boundaries.

Methodology

The study uses 178 labelled MeerLICHT image/mask pairs split at image level into 122 training, 24 validation, and 32 test images, avoiding same-image leakage. Images are tiled into 528×528 patches. The U-Net is reimplemented in PyTorch from the released Keras model configuration, preserving average pooling, LeakyReLU activations, in-block dropout, Keras-style initialisation, and a combined binary cross-entropy plus squared-Dice loss.

Hyperparameters are selected by validation-only Optuna search, the top trials are retrained across five seeds, and the final threshold is chosen from a validation precision–recall sweep before test metrics are inspected. Hough post-processing is evaluated on a full-coverage tiled canvas that reproduces the parity canvas pixel-for-pixel, with separate precision and completeness caveats.

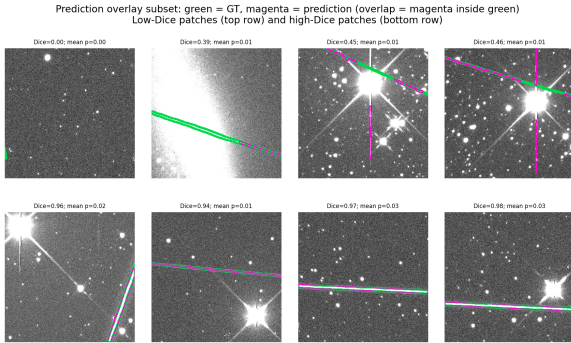


Figure 4: Diagnostic test-patch predictions for the locked U-Net: ground truth in green, prediction in magenta, and overlap inside the green band.

Extensions & Transfer

Follow-up experiments test whether the residual can be moved by model or inference changes. A classifier-gated detector shifts precision by only +1.2 percentage points, consistent with the 17.5% inter-patch upper bound. An attention-gated U-Net at matched capacity is a null result, cDice shows no hidden topology gain, hard-negative mining adds only +0.007 precision, and ensembling with test-time augmentation sits within seed noise.

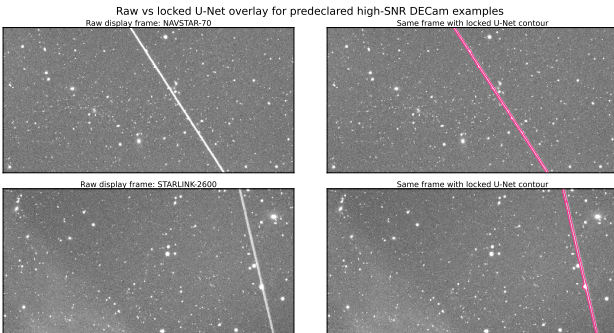


Figure 5: Cold-domain DECam examples: raw measured-streak frames and the locked MeerLICHT-trained U-Net contour. This is a qualitative transfer illustration, not a benchmark.

Recommendation & Impact

The practical result is a high-completeness trail masker whose remaining strict-precision gap is largely boundary-near. The reporting standard should be strict precision/recall plus a boundary-tolerant operating point and an explicit mask-width convention. The strongest remaining check is a multi-annotator annotation audit, which would establish inter-annotator agreement and test how much of the strict precision gap reflects mask granularity rather than detector error. A labelled cross-instrument test is the next generalisation check; the current DECam application is qualitative only.

The U-Net plus Hough pipeline closely reproduces the target detector’s recall and Hough completeness. The tested capacity, training-protocol and post-processing changes do not materially reduce the remaining strict-precision residual. For operational masking, the evidence points to reporting strict scores alongside boundary-tolerant scores and annotation conventions rather than prioritising another architecture tweak.

References

- [1] F. Stoppa et al., “Automated detection of satellite trails in ground-based observations using U-Net and Hough transform,” *Astronomy & Astrophysics*, 692, A199, 2024.
- [2] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” *MICCAI*, 2015.
- [3] J. Matas, C. Galambos, and J. Kittler, “Robust detection of lines using the progressive probabilistic Hough transform,” *Computer Vision and Image Understanding*, 78(1), 2000.

Declaration of Use of Autogeneration Tools

I have used Codex (ChatGPT) and Claude Code to support me across this work, primarily in the following areas:

- Providing advice and usage support for libraries like PyTorch and PIL.
- Generating boilerplate code for plotting, testing and scripting.
- Generating function docstrings.
- Reviewing and auditing project structure for completeness, consistency, accuracy and best practices in software engineering.